

东昆仑西段夏日哈木地区鄂拉山组安山岩对古特提斯洋演化的制约^{*}

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Abstract The volcanic rocks of the Elashan Formation exposed in the Xiarihamu area, western segment of East Kunlun Orogenic Belt, are of great significance for studying the evolution of the Paleo-Tethys Ocean. Detailed zircon U-Pb, Hf isotope, whole-rock geochemistry and Sm-Nd isotope analyses upon the andesites in Xiarihamu area are conducted to determine their formation ages and petrogenesis, and discuss the tectonic significance. The results show that the zircon $^{206}\text{Pb}/^{238}\text{U}$ weighted average age of the andesites is $223 \pm 2\text{Ma}$, belonging to the Late Triassic. The andesites have low SiO_2 (59.45% ~ 63.10%) contents and $\text{Mg}^\#$ values (28 ~ 51), high Al_2O_3 (16.19% ~ 17.66%) and K_2O (1.42% ~ 3.50%) contents, suggesting a weak peraluminous, high-potassium calcium alkaline series. Furthermore, they are enriched in LREE and LILE contents, and depleted in HFSE, with whole-rock $\varepsilon_{\text{Nd}}(t)$ and zircon $\varepsilon_{\text{Hf}}(t)$ values of $-4.3 \sim -3.1$ and $-3.4 \sim +1.1$, respectively. All these characteristics show that the andesites were derived from partial melt of an ancient lower crust which was also influenced by underplating of an enriched lithospheric mantle magma. Based on regional geological data, it is suggested that the andesites in Xiarihamu area are related to Paleo-Tethys evolution and formed in a post-collisional extensional setting instead of a compressive setting. The results of this paper also indicate that it should be careful to identify a subduction environment by using the island arc characteristics of igneous rocks in the orogenic belt.

Key words Andesite; Geochemistry; Sm-Nd isotope; Xiarihamu; East Kunlun

摘 要 东昆仑西段夏日哈木地区出露一套鄂拉山组火山岩,其成因对古特提斯洋演化过程具有重要的指示意义。通过对其进行锆石 U-Pb 年代学、Hf 同位素和全岩主、微量元素、Sm-Nd 同位素等研究确定其形成时代、岩石成因及其构造意义。安山岩中锆石的 $^{206}\text{Pb}/^{238}\text{U}$ 加权平均年龄为 $223 \pm 2\text{Ma}$,属于晚三叠世。岩石具有较低的 SiO_2 (59.45% ~ 63.10%) 和 $\text{Mg}^\#$ 值 (28 ~ 51)、较高的 Al_2O_3 (16.19% ~ 17.66%) 和 K_2O 含量 (1.42% ~ 3.50%),表现为弱过铝高钾钙碱性系列。岩石具有富集轻稀土和大离子亲石元素,亏损高场强元素的特征,结合全岩 $\varepsilon_{\text{Nd}}(t)$ 值 ($-4.3 \sim -3.1$) 和锆石 $\varepsilon_{\text{Hf}}(t)$ 值 ($-3.4 \sim +1.1$),表明其可能来源于受富集地幔熔体底侵的古老下地壳部分熔融。结合区域地质资料,认为这些安山岩形成于碰撞后伸展环境,而古特提斯洋在晚三叠世已关闭。本文研究结果也说明在造山带研究中需谨慎利用火成岩的岛弧性质来判断其俯冲环境这

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一构造属性。

关键词 安山岩; 地球化学; Sm-Nd 同位素; 夏日哈木; 东昆仑
中图法分类号 P588.144; P597.3

东昆仑造山带是青藏高原东北部一条规模巨大的复合岩浆构造带,具有多陆块多旋回的构造属性,主要经历了新元古代-中生代初期与原-古特提斯洋演化有关的陆块汇聚和增生等构造作用以及中生代陆内构造发展的演化过程(姜春发等, 1992; 许志琴等, 2006)。

东昆仑造山带中木孜塔格-布青山-阿尼玛卿蛇绿岩作为古特提斯洋的残片,记录了古特提斯洋的形成与扩张历史(Chen *et al.*, 2001; 兰朝利等, 2002; Bian *et al.*, 2004; Yang *et al.*, 2009; 刘战庆等, 2011),但洋盆打开和关闭的精确时限仍存在不同认识。如:通常认为古特提斯洋在晚石炭世处于扩张状态(Bian *et al.*, 2004; Yang *et al.*, 2009; 刘战庆等, 2011),近年来也有观点认为古特提斯洋在早古生代和晚古生代之交已经开始打开(Dong *et al.*, 2020; 李文渊等, 2022a, b)。古特提斯洋开始俯冲消减发生在早-中二叠世(Chen *et al.*, 2001; Yang *et al.*, 2009),以形成巨量岛弧岩浆岩为特征(莫宣学等, 2007)。而对于洋盆闭合的时限则存在晚二叠世(殷鸿福和张克信, 1997; Pan *et al.*, 2012)、早三叠世(Huang *et al.*, 2014)、中三叠世(郭正府等, 1998; Xia *et al.*, 2014; Xiong *et al.*, 2014)和晚三叠世之后(Ding *et al.*, 2014; Wu *et al.*, 2019)等多种观点,这种分歧严重制约了对东昆仑晚古生代期间构造演化的认识。产生这一分歧的主要原因是东昆仑造山带印支期火山岩的成因和构造属性认识不同,如部分学者认为其来源于古老陆壳部分熔融的产物(封铨等, 2022),也有学者认为其可能为新生下地壳部分熔融形成(Zhu *et al.*, 2022),还有学者认为其形成与古特提斯洋洋壳部分熔融有关(Huang *et al.*, 2014; Shao *et al.*, 2017, 2021)。另外,东昆仑大量金、银、铜、铅、锌等多金属矿床的形成也与三叠纪的岩浆作用有关,如东昆仑东段哈日扎、那更、赛什堂等地陆续发现与晚三叠世鄂拉山组陆相火山岩有关的铅、银多金属矿床(Feng *et al.*, 2017; Zhang *et al.*, 2017; Chen *et al.*, 2022),表明该地层具有重要的成矿潜力。因此,对三叠纪火山岩特别是鄂拉山组火山岩的研究具有重要的理论意义和经济价值。

本次研究选取东昆仑西段夏日哈木地区广泛出露的一套印支期安山岩作为研究对象,通过对其进行详细的锆石 U-Pb 年代学、Hf 同位素和全岩主、微量元素、Sm-Nd 同位素等研究,探讨其时代归属、岩石成因及其对区域构造演化的意义,为东昆仑西段古特提斯洋演化提供新的证据。

1 区域地质背景

东昆仑造山带位于青藏高原的东北部(图 1),北部以柴达木盆地南缘断裂为界与柴达木盆地分隔,南邻布青山-阿

尼玛卿构造混杂岩带及巴颜喀拉造山带,西部以阿尔金断裂为界与西昆仑相连,东部大致以温泉断裂与秦岭相接(李荣社等, 2007)。东昆仑造山带以东昆中断裂带为界分为东昆北地体和东昆南地体(姜春发等, 1992)。东昆北地体以出露大面积中-新元古代变质基底金水口群(Liu *et al.*, 2005; He *et al.*, 2016)和加里东期-印支期的侵入岩(莫宣学等, 2007)为特征。金水口群的岩性以片麻岩-斜长角闪岩-大理岩为主,后期经历了新元古代(He *et al.*, 2016)和早古生代(张建新等, 2003; Liu *et al.*, 2005; 李怀坤等, 2006; Wang *et al.*, 2022b)变质热事件,变质变形强烈,局部可达麻粒岩-榴辉岩相(Meng *et al.*, 2013; 孟繁聪等, 2015; Song *et al.*, 2018; Bi *et al.*, 2022)。该区还分布大量加里东期-印支期的侵入岩,其中加里东期侵入岩主要出露泥盆纪花岗岩,少量的闪长岩和辉长岩(Duan *et al.*, 2020),印支期侵入岩主要出露晚二叠世-三叠纪花岗岩、基性岩体及基性岩墙群(Dong *et al.*, 2018)。

上述岩石的形成主要与其经历了早古生代原特提斯洋和晚古生代-早中生代古特提斯洋两个不同构造旋回的演化过程有关(莫宣学等, 2007)。已有多位学者阐述了东昆仑造山带多旋回、多板块、多期次拼贴的构造演化历史(姜春发等, 1992; 许志琴等, 2006)。东昆仑早古生代洋盆在早寒武世之前打开和扩张(Yang *et al.*, 1996; Bian *et al.*, 2004; 祁晓鹏等, 2016; Li *et al.*, 2019, 2021),早奥陶-中志留世原特提斯洋持续俯冲消减,多数学者认为原特提斯洋在 ~430Ma 闭合(Meng *et al.*, 2013; 刘彬等, 2013),并且在早泥盆世(~420Ma)构造体制由挤压转变为伸展(陆露等, 2010; Song *et al.*, 2016)。晚石炭世古特提斯洋开始扩张(340 ~ 333Ma, Yang *et al.*, 2009; 刘战庆等, 2011),中二叠世-早三叠世(270 ~ 240Ma)持续俯冲,主要形成钙碱性闪长岩-石英闪长岩,240Ma 之后主要形成富钾钙碱性含暗色微粒包体为特征的花岗岩-花岗闪长岩(陈国超等, 2020; Shao *et al.*, 2021)。

2 研究区地质及火山岩特征

研究区位于东昆仑造山带西段东昆北地体(图 1),地理位置处于青海省格尔木市西部夏日哈木地区。研究区南部主要出露中-新元古代金水口群片麻岩,夹有大理岩、斜长角闪岩(李爽等, 2021)和榴辉岩透镜体(祁生胜等, 2014; 范亚洲等, 2018),新元古代花岗岩侵入到片麻岩中(王冠等, 2016)。此外,侵入到片麻岩中的还有镁铁-超镁铁岩,其形成时代为 430 ~ 400Ma,这些超基性岩体中赋存有超大型镍-钴硫化物矿床(王冠等, 2014a; Li *et al.*, 2015; 姜常义等,

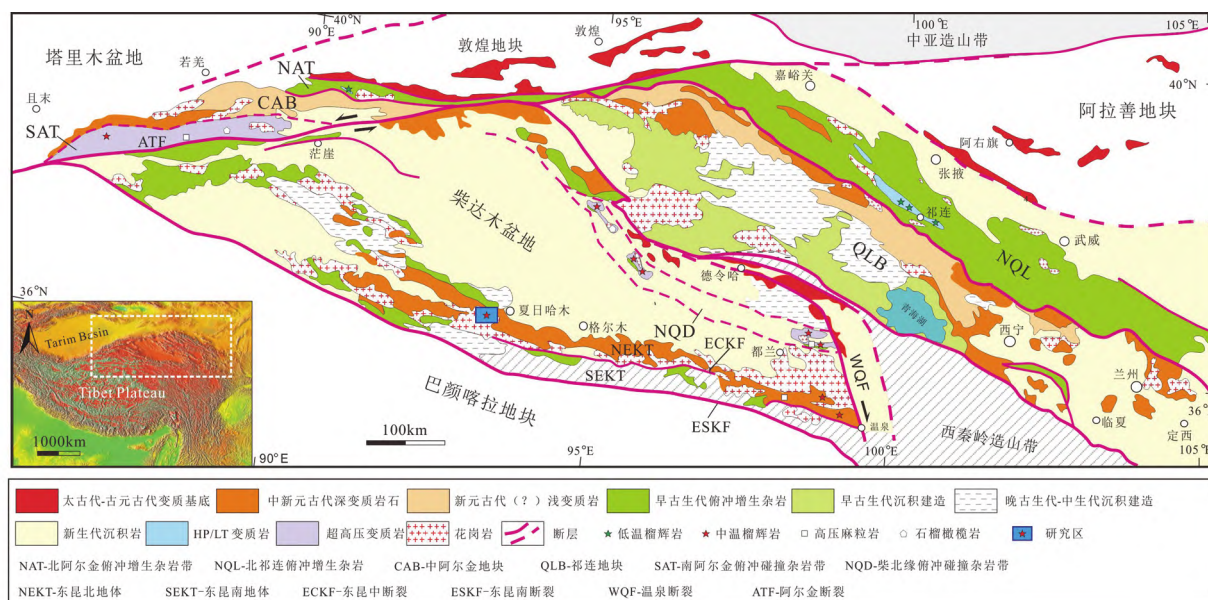


图1 东昆仑造山带及邻区区域构造格架简图(据张建新等,2015 修改)

Fig. 1 Tectonic framework of the East Kunlun Orogenic Belt and adjacent region (modified after Zhang *et al.* 2015)

2015; Song *et al.*, 2016; Zhang *et al.*, 2019), 在夏日哈木矿区北部还出露早泥盆世正长花岗岩(王冠等, 2013)。研究区北部主要出露奥陶系纳赤台群长石砂岩、玄武岩和白云岩, 大面积鄂拉山组火山岩地层, 该地层角度不整合覆盖在奥陶系以及石炭系地层之上, 局部与中泥盆世正长花岗岩断层接触(图2)。区内还出露少量印支期闪长岩(王冠等, 2014b)、石英闪长岩、二长花岗岩和花岗闪长岩等侵入岩。

区内火山碎屑岩与火山熔岩呈互层状产出, 并以火山碎屑岩为主, 可见火山角砾岩、凝灰岩等; 熔岩次之, 主要为中-酸性岩, 包括安山岩、英安岩以及流纹岩(赵明福等, 2018), 未见基性岩出露, 岩石蚀变较弱, 局部发育沉积岩夹层(图3a)。本次采集样品皆为安山岩, 西部样品 K17-5-8 采样坐标: $36^{\circ}38'12''N, 93^{\circ}19'16''E$; 东部样品 K17-2-2 采样坐标: $36^{\circ}32'05''N, 93^{\circ}32'44''E$ 。野外可见同期花岗闪长岩与该安山岩直接接触(图3b)。安山岩蚀变较弱, 手标本主要呈灰黑色, 风化后呈铁锈色。显微镜下东、西段样品略有区别: 西段样品斑晶主要由斜长石和单斜辉石组成, 少量绿泥石化角闪石(图3c), 其中斜长石粒径 $0.5 \sim 1\text{mm}$, 含量约 30%, 单斜辉石粒径 $0.2 \sim 0.6\text{mm}$, 含量约 15%, 副矿物见少量磁铁矿; 东段样品斑晶主要以具有环带特征的斜长石和绿泥石化角闪石为主(图3d), 其中斜长石粒径 $0.5 \sim 1.5\text{mm}$, 含量约 35%, 绿泥石化角闪石粒径 $0.2 \sim 0.5\text{mm}$, 含量约 15%。斑晶矿物组成表明西段样品基性程度更高。

3 测试方法

3.1 全岩主、微量元素

全岩主、微量元素在自然资源部国家地质实验测试中心

完成。主量元素通过 XRF(X 荧光光谱仪 3080E) 方法测试, 分析精度 5%。微量元素和稀土元素(REE) 通过等离子质谱仪(ICP-MS-Excel) 分析, 其中 Nb、Ta、Zr 和 Hf 用碱溶法和沉淀酸提取, 用等离子质谱法进行测定, 微量元素和稀土元素含量大于 10×10^{-6} 的元素测试精度约为 5%, 而小于 10×10^{-6} 的元素精度约为 10%。

3.2 Sm-Nd 同位素组成

Sm-Nd 同位素测试在中国科学技术大学地球与空间科学学院固体同位素地球化学实验室完成。样品的化学分离纯化在净化实验室完成, 比值测定采用同位素稀释法在热电质谱计 MAT262 上进行。Nd 同位素比值测定采用 $^{146}\text{Nd}/^{144}\text{Nd} = 0.7219$ 进行质量分馏标准化校正, 化学流程和同位素比值测定见 Chen *et al.* (2007)。

3.3 锆石 LA-ICP-MS U-Pb 测年

测年样品锆石的挑选在河北省地质矿产局廊坊区调院实验室利用标准重矿物分离技术分选完成。样品经常规方法破碎研磨, 淘洗后用重液分选, 在双目镜下挑选锆石颗粒, 选择晶形完好并且纯净透明的锆石, 将锆石和标准锆石 TEM 一起粘在玻璃板上, 对其进行反射光、透射光和阴极发光(CL) 图像照相以及 LA-ICP-MS 分析。

锆石 U-Pb 同位素测试在天津地质矿产研究所完成, 所用仪器为 Neptune 多接收电感耦合等离子体质谱仪和 193nm 激光取样系统(LA-MC-ICP-MS)。选择锆石中的合适区域, 利用 193nm FX 激光器对锆石进行剥蚀, 束斑直径为 $35\mu\text{m}$, 激光能量密度为 $13 \sim 14\text{J}/\text{cm}^2$, 频率为 $8 \sim 10\text{Hz}$ 。所用锆石标样为 TEMORA 标准锆石, 数据处理采用 ICP-MS DataCal 程

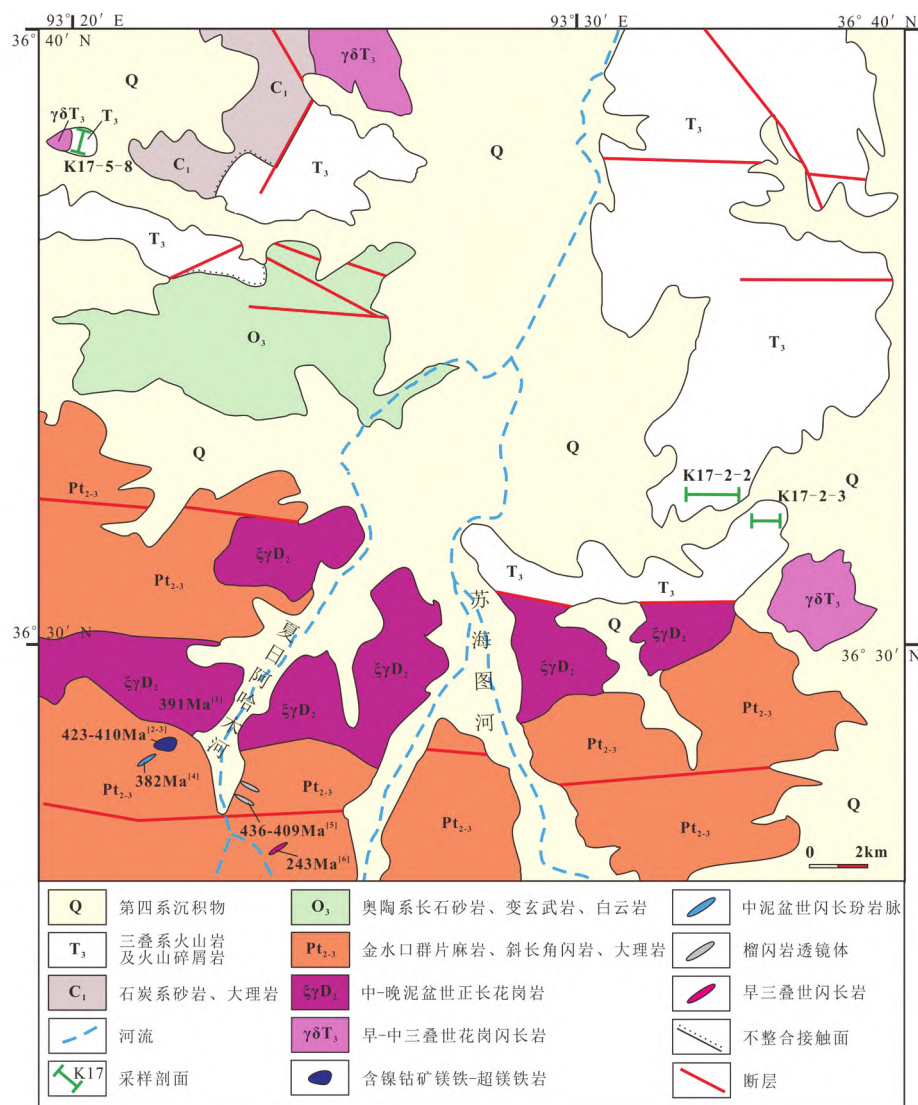


图2 夏日哈木地区地质简图(据王冠等, 2014a; 赵明福等, 2018 修改)

年龄数据来源 [1] 王冠等, 2013; [2] 王冠等, 2014a; [3] Song et al., 2016; [4] 奥琮等, 2014; [5] 张照伟等, 2017; [6] 王冠等, 2014b

Fig. 2 Sketch geological map of the Xiarihamu area (modified after Wang et al., 2014a; Zhao et al., 2018)

Age data source [1] Wang et al., 2013; [2] Wang et al., 2014a; [3] Song et al., 2016; [4] Ao et al., 2014; [5] Zhang et al., 2017; [6] Wang et al., 2014b

序(Liu et al., 2010)和ISOPLLOT程序(Ludwig, 2003)进行分析和作图,采用 ^{208}Pb 对普通铅进行校正,利用NIST612作为外标计算锆石样品的Pb、U含量(李怀坤等, 2009)。采用 $^{206}\text{Pb}/^{238}\text{U}$ 年龄的加权平均年龄值,数据误差为 1σ ,加权平均年龄值误差为 1σ 。

3.4 锆石 Lu-Hf 同位素

锆石 Lu-Hf 同位素的测定在天津地质矿产研究所完成,测试仪器为 Neptune 多接收等离子质谱和 Newwave UP213 激光剥蚀系统(LA-MC-ICP-MS),实验中以 He 作为载气,剥蚀束斑直径采用 $35\mu\text{m}$,参考物质使用锆石国际标样 GJ-1,分析点近似位于锆石 U-Pb 定年分析点位置。仪器运行操作条件

及详细分析流程见耿建珍等(2011)。分析过程中标样 GJ1 的 $^{176}\text{Hf}/^{177}\text{Hf}$ 测试加权平均值为 0.282015 ± 10 (2σ , $n = 10$)。

4 分析结果

4.1 全岩地球化学

本次研究选取 13 块新鲜岩石样品进行全岩地球化学测试,测试结果见表 1。样品 SiO_2 含量为 $59.45\% \sim 63.10\%$, TiO_2 含量介于 $0.72\% \sim 0.80\%$ 之间, K_2O 含量在 $1.42\% \sim 3.50\%$, 平均 2.97% , Na_2O 含量在 $2.51\% \sim 3.70\%$, $\text{Mg}^\#$ 为 $28 \sim 51$, 平均 45 。样品的 A/CNK 在 $1.2 \sim 2.0$ 之间,且 Al_2O_3

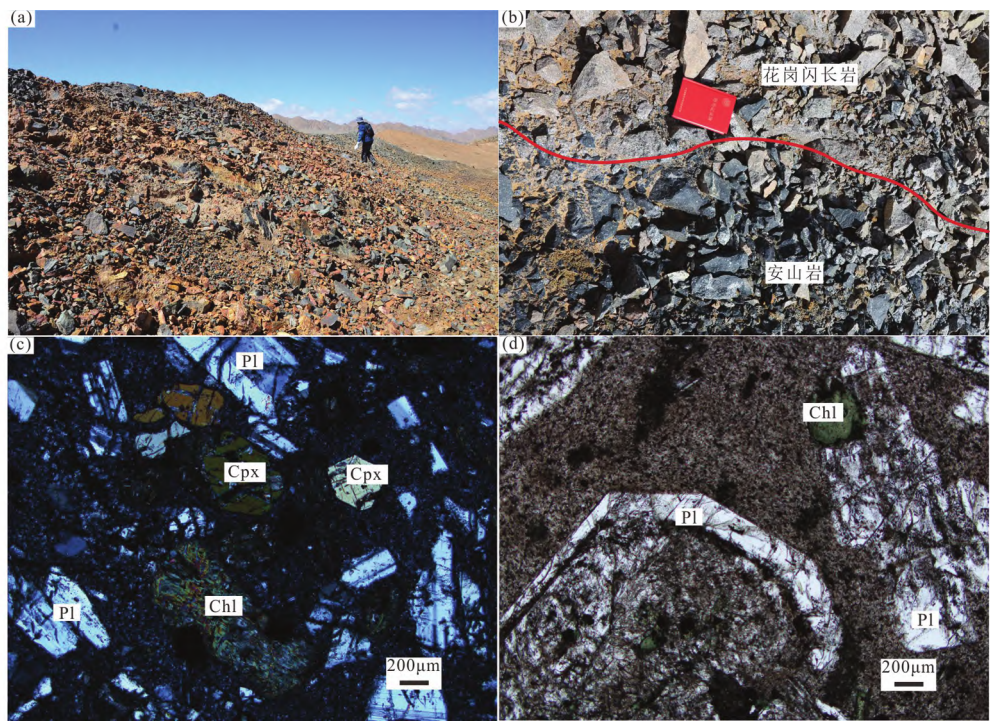


图3 夏日哈木地区鄂拉山组安山岩特征
(a) 安山岩野外露头; (b) 安山岩与花岗闪长岩接触边界; (c) 安山岩的斜长石和单斜辉石斑晶 (K17-5-8.3, 正交偏光); (d) 安山岩中的斜长石具环带特征, 角闪石发生绿泥石化 (K17-2-2.2, 正交偏光). Pl-斜长石; Cpx-单斜辉石; Chl-绿泥石

Fig. 3 Features of andesites of the Elashan Formation in Xiarihamu area
(a) outcrop of andesites; (b) the contact boundary between the andesite and granodiorite; (c) phenocrysts of plagioclase and clinopyroxene (K17-5-8.3, cross-polarized light); (d) zoned plagioclase and amphibole phenocryst of andesites (K17-2-2.2, cross-polarized light). Pl-plagioclase; Cpx-clinopyroxene; Chl-chlorite

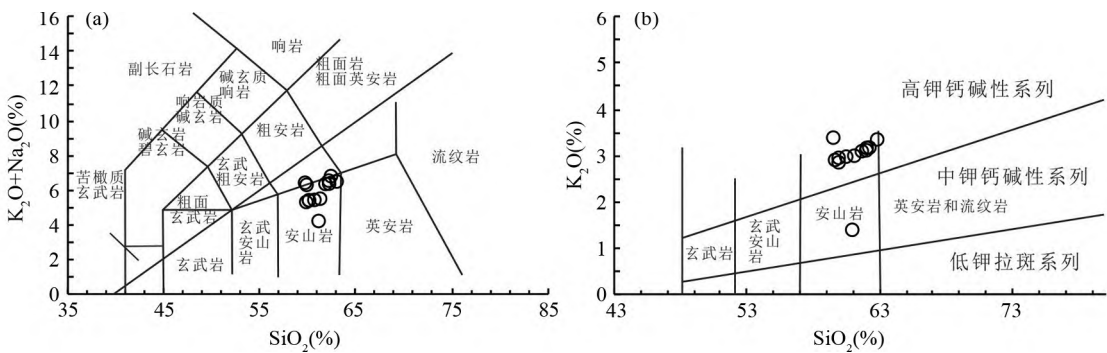


图4 夏日哈木安山岩地球化学分类
(a) 硅碱图 (据 Le Bas *et al.*, 1986); (b) SiO₂-K₂O 图 (据 Peccerillo and Taylor, 1976)

Fig. 4 Chemical classifications of the Xiarihamu andesites
(a) SiO₂ vs. K₂O + Na₂O diagram (after Le Bas *et al.*, 1986); (b) SiO₂ vs. K₂O diagram (after Peccerillo and Taylor, 1976)

含量大于 K₂O、Na₂O 和 CaO 含量之和,反映岩石具有弱铝饱和特征。在火山岩硅碱图中,样品的投点基本落入安山岩区 (图 4a); 在 SiO₂-K₂O 图中,除一个样品落入中钾区域,其余样品均落入高钾系列区域 (图 4b)。

样品稀土总量较高,介于 141.9 × 10⁻⁶ ~ 189.9 × 10⁻⁶ 之间。(La/Yb)_N = 7.71 ~ 10.54,轻稀土富集,重稀土亏损。δEu 介于 0.65 ~ 0.75 之间,具弱的负铕异常。球粒陨石标准

化稀土元素配分曲线呈向右缓倾的轻稀土富集型 (图 5a),但重稀土分馏不明显,具相对平坦的型式,钕具弱负异常特征,各稀土曲线近于平行。安山岩大离子亲石元素 K、Rb、Ba 的含量较高,Rb 含量为 54.6 × 10⁻⁶ ~ 178.0 × 10⁻⁶、Ba 为 462 × 10⁻⁶ ~ 840 × 10⁻⁶。Sr 含量为 239 × 10⁻⁶ ~ 793 × 10⁻⁶。高场强元素方面,Th 含量为 10.8 × 10⁻⁶ ~ 19.5 × 10⁻⁶,Hf 含量为 4.96 × 10⁻⁶ ~ 7.57 × 10⁻⁶,U 含量为 2.5 × 10⁻⁶ ~ 4.3 ×

表1 夏日哈木安山岩全岩主量(wt%)、微量($\times 10^{-6}$)元素含量Table 1 Major (wt%) and trace ($\times 10^{-6}$) elements analysis results of the Xiarihamu andesites

样品号	K17- 5-8.2	K17- 5-8.3	K17- 5-8.4	K17- 5-8.5	K17- 5-8.6	K17- 5-8.7	K17- 5-8.8	K17- 2-2.2	K17- 2-2.3	K17- 2-3.1	K17- 2-3.2	K17- 2-3.3	K17- 2-3.4
SiO ₂	59.45	60.53	59.62	59.59	59.91	59.90	61.20	62.21	62.45	63.10	61.82	62.23	61.02
TiO ₂	0.75	0.75	0.72	0.80	0.79	0.79	0.76	0.73	0.75	0.74	0.75	0.73	0.72
Al ₂ O ₃	16.40	16.40	16.19	16.59	16.56	16.35	16.68	17.43	17.20	17.37	17.61	17.66	16.98
Fe ₂ O ₃	1.36	1.77	1.76	1.15	1.38	1.50	1.96	1.66	1.77	1.75	1.82	2.55	2.23
FeO	4.11	4.08	3.79	4.80	4.51	4.47	3.79	2.53	2.46	1.74	2.35	1.71	2.17
MnO	0.11	0.11	0.10	0.11	0.11	0.12	0.10	0.09	0.09	0.12	0.08	0.07	0.18
MgO	2.81	2.80	2.47	3.19	3.17	3.34	2.67	1.88	1.89	1.35	1.63	1.62	0.90
CaO	4.27	5.54	4.73	5.89	5.88	5.35	5.55	4.41	4.07	4.47	4.86	5.11	6.92
Na ₂ O	3.12	2.56	2.86	2.51	2.64	2.58	2.60	3.51	3.70	3.23	3.35	3.30	2.91
K ₂ O	3.36	2.96	3.50	2.89	2.84	2.93	2.98	3.09	3.15	3.32	3.07	3.14	1.42
P ₂ O ₅	0.14	0.14	0.13	0.15	0.15	0.15	0.14	0.21	0.21	0.21	0.22	0.21	0.21
LOI	3.07	1.43	2.94	1.71	1.65	2.07	1.26	1.93	1.58	2.06	1.76	1.68	4.20
Total	98.95	99.07	98.81	99.38	99.59	99.55	99.69	99.68	99.32	99.46	99.32	100.01	99.86
La	32.00	32.10	33.80	28.90	30.40	29.70	35.20	40.00	38.70	37.40	41.10	38.70	39.50
Ce	68.60	58.80	62.20	62.80	50.90	59.80	60.50	67.40	67.20	63.50	69.50	65.00	69.30
Pr	7.34	7.37	7.93	6.97	7.09	7.27	8.18	9.24	9.11	8.64	9.43	8.58	8.91
Nd	26.40	31.30	33.50	26.40	29.40	26.70	34.00	41.10	40.40	38.40	42.50	37.20	38.80
Sm	5.37	5.72	5.95	5.44	5.50	5.56	6.24	6.81	7.08	6.54	7.07	6.41	6.74
Eu	1.17	1.23	1.28	1.15	1.18	1.17	1.30	1.54	1.43	1.43	1.53	1.46	1.55
Gd	4.86	5.13	5.27	4.81	5.00	5.07	5.72	5.69	5.76	5.59	5.63	5.48	5.54
Tb	0.83	0.86	0.89	0.80	0.79	0.84	0.92	0.93	0.90	0.88	0.90	0.83	0.88
Dy	4.81	5.03	5.03	4.57	4.65	4.85	5.37	5.09	4.85	4.84	4.86	4.50	4.76
Ho	0.89	0.98	1.01	0.90	0.91	0.94	1.06	1.03	1.00	1.00	1.02	0.96	0.99
Er	2.68	2.79	2.89	2.59	2.63	2.74	2.98	2.78	2.71	2.78	2.70	2.61	2.68
Tm	0.40	0.41	0.44	0.40	0.41	0.41	0.46	0.45	0.43	0.43	0.43	0.41	0.42
Yb	2.73	2.71	2.99	2.69	2.64	2.73	2.97	2.94	2.79	2.74	2.83	2.70	2.69
Lu	0.37	0.38	0.42	0.36	0.39	0.37	0.42	0.42	0.40	0.40	0.36	0.39	0.39
ΣREE	158.5	154.8	163.6	148.8	141.9	148.2	165.3	185.4	182.8	174.6	189.9	175.2	183.2
Y	26.7	28.3	28.1	25.7	24.6	25.9	29.9	28.8	28.8	31.6	29.8	26.6	28.2
Rb	130.0	104.0	138.0	92.8	97.5	106.0	118.0	92.7	91.0	92.2	76.1	83.0	54.6
Sr	341.0	315.0	324.0	319.0	328.0	341.0	330.0	544.0	586.0	564.0	576.0	582.0	793.0
Ba	618.0	530.0	631.0	468.0	462.0	539.0	588.0	767.0	790.0	814.0	786.0	759.0	840.0
Nb	9.79	9.73	10.10	8.70	8.65	9.39	10.90	11.90	12.40	12.40	12.30	11.80	12.70
Ta	0.77	0.79	0.81	0.67	0.73	0.73	0.81	0.76	0.76	0.71	0.74	0.70	0.77
Zr	177.0	179.0	182.0	166.0	167.0	172.0	203.0	245.0	237.0	251.0	245.0	226.0	244.0
Hf	5.35	5.48	5.72	4.96	5.19	5.41	5.91	6.75	7.01	6.78	6.71	6.87	6.95
Th	14.10	14.10	14.60	11.80	12.00	12.50	15.10	11.10	11.30	10.80	11.00	11.00	11.50
U	3.18	3.70	3.57	2.98	2.86	2.99	3.98	2.65	2.73	2.51	2.64	2.72	2.99
V	112.0	115.0	112.0	117.0	115.0	123.0	128.0	87.60	85.70	83.30	87.90	85.70	86.30
Co	14.30	14.40	14.20	16.50	14.20	15.30	15.40	8.22	8.23	14.40	7.99	7.95	6.72
Ni	8.19	9.09	8.50	9.57	10.40	9.14	9.44	5.97	4.73	6.55	4.58	4.62	4.95
Sc	16.60	17.30	17.20	16.50	16.50	17.30	18.20	10.60	10.70	10.90	11.00	10.50	9.19
(La/Yb) _N	9.76	9.95	9.80	10.42	10.29	10.54	8.41	8.50	8.11	7.71	8.26	7.81	8.51
δEu	0.74	0.66	0.71	0.72	0.74	0.75	0.69	0.68	0.69	0.67	0.68	0.66	0.65

注: $\delta\text{Eu} = 2 \times \text{Eu} / (\text{Sm} + \text{Gd})$

表 2 夏日哈木安山岩 Sm-Nd 同位素数据

Table 2 Sm-Nd isotopic compositions of the Xiarihamu andesites

样品号	Sm($\times 10^{-6}$)	Nd($\times 10^{-6}$)	$^{147}\text{Sm}/^{144}\text{Nd}$	$^{143}\text{Nd}/^{144}\text{Nd}$	$\pm 2\sigma$	$f_{\text{Sm}/\text{Nd}}$	$\varepsilon_{\text{Nd}}(0)$	$\varepsilon_{\text{Nd}}(t)$	$t_{\text{DM1}}(\text{Ma})$	$t_{\text{DM2}}(\text{Ma})$
K17-5-8.2	5.18	23.8	0.1315	0.512384	0.000009	-0.3315	-5.0	-3.1	1418	1244
K17-5-8.3	5.35	24.6	0.1312	0.512355	0.00001	-0.3330	-5.5	-3.7	1466	1290
K17-5-8.4	5.05	23	0.1325	0.512378	0.00001	-0.3264	-5.1	-3.2	1447	1255
K17-5-8.5	5.32	24.6	0.1305	0.512326	0.00001	-0.3366	-6.1	-4.2	1507	1336
K17-5-8.6	5.3	24.1	0.133	0.512334	0.00001	-0.3238	-5.9	-4.1	1538	1326
K17-5-8.7	5.51	25	0.1333	0.512355	0.000009	-0.3223	-5.5	-3.7	1505	1293
K17-5-8.8	5.33	24.7	0.1307	0.512319	0.000007	-0.3355	-6.2	-4.3	1523	1347

注: $t = 223\text{Ma}$ (本文测试)

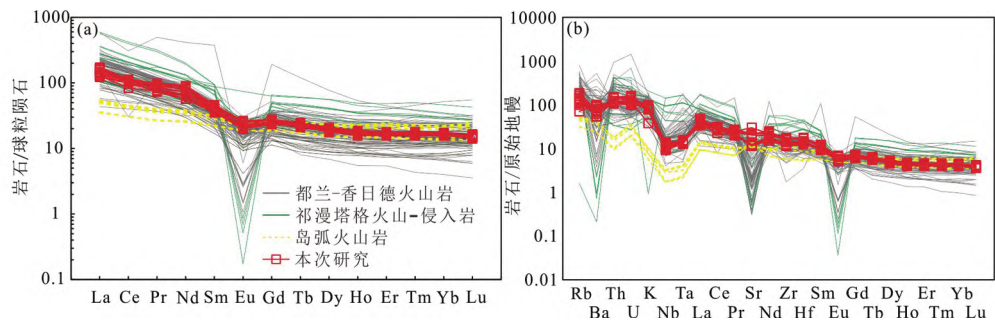


图 5 夏日哈木安山岩球粒陨石标准化稀土元素配分曲线(a)和原始地幔标准化微量元素蛛网图(b) (标准化值据 Sun and McDonough, 1989)

数据来源: 都兰-香日德火山岩据 Li *et al.* (2015) , Fan *et al.* (2022) , Chen *et al.* (2022) , Zhu *et al.* (2022) ; 祁漫塔格火山-侵入岩据 Hu *et al.* (2016) , Yin *et al.* (2017) ; 岛弧火山岩据 Tamura *et al.* (2014)

Fig. 5 Chondrite-normalized REE patterns (a) and primitive mantle-normalized trace element spider diagrams (b) of the Xiarihamu andesites (normalization values after Sun and McDonough, 1989)

Data Sources: Dulan-Xiangride volcanic rocks from Li *et al.* (2015) , Fan *et al.* (2022) , Chen *et al.* (2022) and Zhu *et al.* (2022) ; Qimantag volcanic and intrusive rocks from Hu *et al.* (2016) and Yin *et al.* (2017) ; Arc volcanic rocks from Tamura *et al.* (2014)

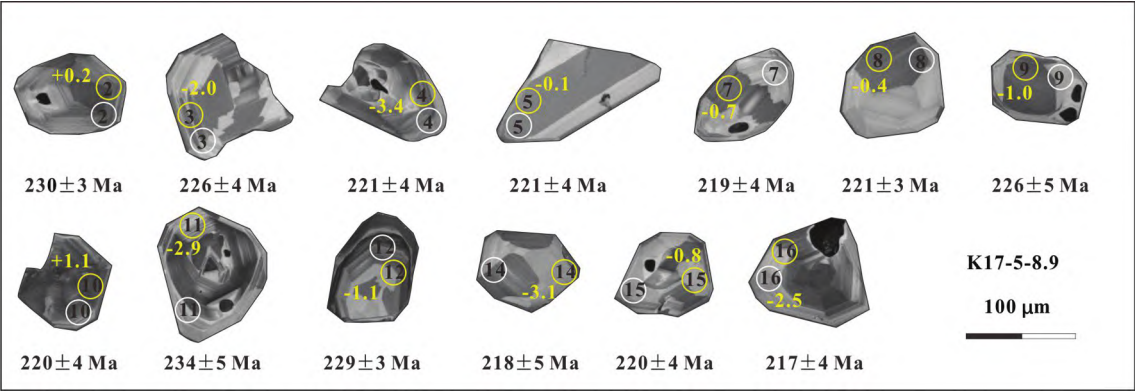


图 6 夏日哈木安山岩代表性锆石阴极发光图片
白色圈为 U-Pb 测点,黄色圈为 Lu-Hf 测点

Fig. 6 Representative cathodoluminescent images of zircons from the Xiarihamu andesites
Yellow circle for Lu-Hf isotope analyses; white circle for U-Pb age analyses

10^{-6} , Zr 为 $166 \times 10^{-6} \sim 252 \times 10^{-6}$, 含量均较高。Nb、Ta、Ti 含量较低, Nb 含量为 $8.7 \times 10^{-6} \sim 13.0 \times 10^{-6}$, Ta 为 $0.67 \times 10^{-6} \sim 0.93 \times 10^{-6}$ 。在原始地幔标准化微量元素蛛网图(图 5b) 中, 表现为强不相容元素 Rb、Ba、U、Th 强烈富集, 中等不相容元素 La、Ce、Nd、Zr、Hf、Sm 等中等富集, Ta、Nb 亏损, 弱不相容元素 Ti、P 亏损, K 富集, 其特征与岛弧火山岩分布型

表 3 夏日哈木安山岩(样品 K17-5-8.9) LA-ICP-MS 锆石 U-Pb 定年分析结果

Table 3 LA-ICP-MS U-Pb analysis results of the zircons from the Xiarihamu andesite (Sample K17-5-8.9)

测点号	含量($\times 10^{-6}$)		同位素比值								年龄(Ma)					
	Pb	U	$\frac{^{238}\text{U}}{^{232}\text{Th}}$	1σ	$\frac{^{206}\text{Pb}}{^{238}\text{U}}$	1σ	$\frac{^{207}\text{Pb}}{^{235}\text{U}}$	1σ	$\frac{^{207}\text{Pb}}{^{206}\text{Pb}}$	1σ	$\frac{^{206}\text{Pb}}{^{238}\text{U}}$	1σ	$\frac{^{207}\text{Pb}}{^{235}\text{U}}$	1σ	$\frac{^{207}\text{Pb}}{^{206}\text{Pb}}$	1σ
1	5.9	123.9	0.8368	0.0011	0.0349	0.0007	0.2892	0.0214	0.0604	0.0046	221	4	258	17	617	166
2	8.7	186.1	0.9816	0.0011	0.0363	0.0005	0.2468	0.0120	0.0495	0.0024	230	3	224	10	169	115
3	5.0	103.3	0.7437	0.0008	0.0357	0.0007	0.2491	0.0149	0.0503	0.0029	226	4	226	12	209	136
4	9.8	214.8	0.8952	0.0010	0.0349	0.0006	0.2638	0.0145	0.0550	0.0031	221	4	238	12	413	160
5	9.0	186.0	0.6562	0.0045	0.0348	0.0006	0.2539	0.0111	0.0527	0.0020	221	4	230	9	317	119
6	10.1	237.8	1.1063	0.0115	0.0325	0.0006	0.2520	0.0142	0.0561	0.0028	206	4	228	11	454	111
7	5.6	122.9	0.9121	0.0008	0.0346	0.0006	0.2694	0.0148	0.0566	0.0030	219	4	242	12	476	117
8	6.9	143.9	0.8243	0.0044	0.0348	0.0005	0.2829	0.0122	0.0590	0.0024	221	3	253	10	565	87
9	8.5	178.5	0.8389	0.0012	0.0357	0.0007	0.2785	0.0154	0.0568	0.0030	226	5	249	12	483	114
10	12.3	263.3	0.8455	0.0005	0.0348	0.0006	0.2461	0.0124	0.0514	0.0023	220	4	223	10	257	108
11	8.8	179.5	0.8713	0.0012	0.0369	0.0007	0.2637	0.0128	0.0515	0.0021	234	5	238	10	265	93
12	20.7	438.2	1.0010	0.0017	0.0362	0.0005	0.2500	0.0086	0.0501	0.0016	229	3	227	7	198	76
13	8.2	180.8	0.8422	0.0016	0.0334	0.0005	0.2343	0.0133	0.0510	0.0029	212	3	214	11	239	133
14	5.1	109.7	0.8486	0.0011	0.0344	0.0008	0.2413	0.0183	0.0515	0.0038	218	5	219	15	265	173
15	11.5	213.8	0.5742	0.0024	0.0347	0.0006	0.2527	0.0103	0.0533	0.0024	220	4	229	8	339	102
16	9.2	193.1	0.7408	0.0013	0.0343	0.0006	0.2457	0.0127	0.0523	0.0028	217	4	223	10	302	122
17	3.9	82.9	0.9980	0.0001	0.0352	0.0008	0.2269	0.0177	0.0472	0.0035	223	5	208	15	58	176
18	10.1	217.2	0.8955	0.0005	0.0348	0.0006	0.2508	0.0111	0.0524	0.0023	220	3	227	9	306	128
19	12.9	259.9	0.6415	0.0064	0.0350	0.0005	0.2395	0.0088	0.0496	0.0017	222	3	218	7	176	75
20	16.7	364.2	0.8245	0.0018	0.0334	0.0005	0.2291	0.0099	0.0499	0.0023	212	3	209	8	191	106

表 4 夏日哈木安山岩(样品号 K17-5-8.9) 锆石 Lu-Hf 同位素数据

Table 4 Zircon Lu-Hf isotopic compositions of the Xiarihamu andesite (Sample K17-5-8.9)

测点号	$\frac{^{176}\text{Yb}}{^{177}\text{Hf}}$	1σ	$\frac{^{176}\text{Lu}}{^{177}\text{Hf}}$	1σ	$\frac{^{176}\text{Hf}}{^{177}\text{Hf}}$	1σ	$\frac{^{176}\text{Hf}}{^{177}\text{Hf}_i}$	$\varepsilon_{\text{Hf}}(0)$	$\varepsilon_{\text{Hf}}(t)$	$t_{\text{DM}}(\text{Ma})$	$t_{\text{DM2}}(\text{Ma})$	$f_{\text{Lu/Hf}}$
1	0.0512	0.0003	0.0012	0.000007	0.282665	0.000023	0.282661	-3.8	1.0	836	1601	-0.96
2	0.0611	0.0002	0.0014	0.000007	0.282644	0.000026	0.282638	-4.5	0.2	870	1671	-0.96
3	0.0355	0.0002	0.0008	0.000003	0.282580	0.000021	0.282577	-6.8	-2.0	946	1866	-0.98
4	0.0376	0.0002	0.0008	0.000003	0.282540	0.000021	0.282537	-8.2	-3.4	1003	1994	-0.98
5	0.0506	0.0001	0.0011	0.000003	0.282636	0.000023	0.282631	-4.8	-0.1	875	1694	-0.97
6	0.0351	0.0002	0.0008	0.000001	0.282560	0.000021	0.282557	-7.5	-2.7	974	1931	-0.98
7	0.0433	0.0006	0.0009	0.000007	0.282618	0.000024	0.282614	-5.5	-0.7	897	1750	-0.97
8	0.0576	0.0003	0.0012	0.000003	0.282627	0.000025	0.282622	-5.1	-0.4	891	1724	-0.96
9	0.0685	0.0005	0.0014	0.000008	0.282611	0.000024	0.282605	-5.7	-1.0	919	1778	-0.96
10	0.0506	0.0013	0.0012	0.000021	0.282668	0.000022	0.282663	-3.7	1.1	831	1591	-0.97
11	0.0503	0.0001	0.0011	0.000005	0.282557	0.000022	0.282552	-7.6	-2.9	988	1945	-0.97
12	0.0332	0.0007	0.0007	0.000010	0.282605	0.000023	0.282602	-5.9	-1.1	911	1789	-0.98
13	0.0257	0.0001	0.0006	0.000005	0.282569	0.000021	0.282567	-7.2	-2.4	957	1900	-0.98
14	0.0475	0.0002	0.0010	0.000008	0.282550	0.000024	0.282546	-7.8	-3.1	994	1965	-0.97
15	0.0380	0.0002	0.0008	0.000003	0.282616	0.000020	0.282612	-5.5	-0.8	897	1754	-0.98
16	0.0589	0.0002	0.0012	0.000003	0.282568	0.000025	0.282563	-7.2	-2.5	975	1912	-0.96

注: $t = 223\text{Ma}$ (本文测试)

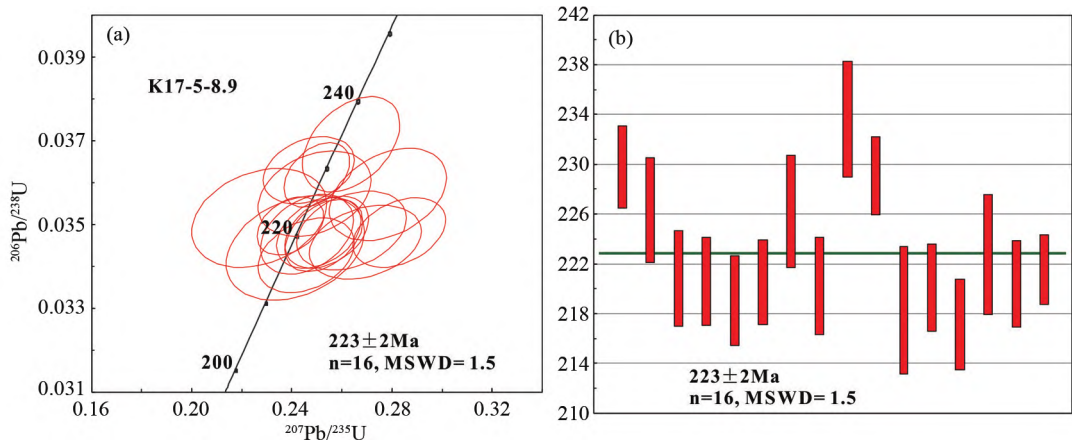


图7 夏日哈木安山岩锆石 U-Pb 同位素年龄谱和图 (a) 和加权平均年龄 (b)

Fig. 7 U-Pb concordia diagrams (a) and weighted-mean $^{206}\text{Pb}/^{238}\text{U}$ ages (b) for zircons from the Xiarihamu andesites

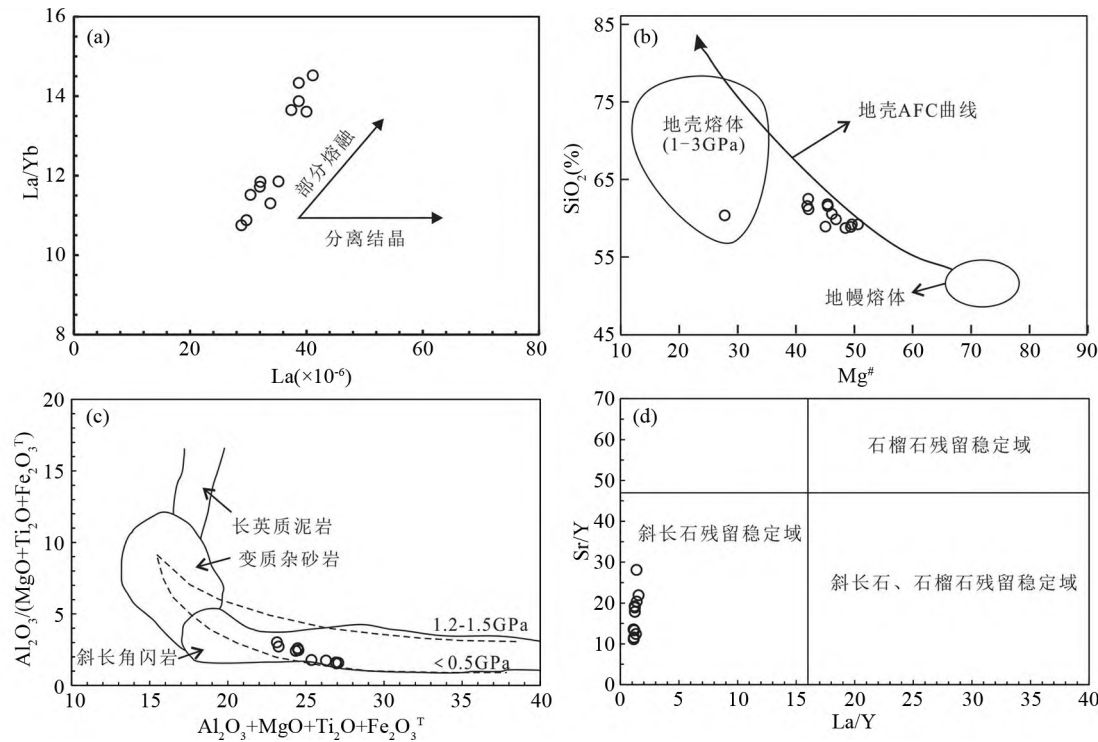


图8 夏日哈木安山岩岩石化学图解

(a) La/Yb - La 图, 矿物分离结晶模拟据 Ersoy and Helvacl (2010); (b) SiO_2 - $\text{Mg}^\#$ 图, 地壳 AFC 曲线据 Stern and Kilian (1996); (c) 主量成分特征图, 高压 (1.2 ~ 1.5 GPa) 和低压 (< 0.5 GPa) 趋势线据 Otamendi *et al.* (1999), 源区数据据 Altherr *et al.* (2000); (d) Sr/Y - La/Y 图, 稳定域范围据 Rapp and Watson (1995)

Fig. 8 Petrochemical diagrams for the Xiarihamu andesite

(a) La/Yb vs. La plot, mineral fractional crystallization simulation from Ersoy and Helvacl (2010); (b) SiO_2 vs. $\text{Mg}^\#$ plot, curves of mantle and crustal AFC from Stern and Kilian (1996); (c) major element plot, high-pressure (1.2 ~ 1.5 GPa) and low-pressure (< 0.5 GPa) from Otamendi *et al.* (1999), source data from Altherr *et al.* (2000); (d) Sr/Y vs. La/Y plot, stability field from Rapp and Watson (1995)

式基本相似(Tamura *et al.* , 2014)。

4.2 Sm-Nd 同位素特征

夏日哈木安山岩的全岩 Sm-Nd 同位素组成见表 2。安

山岩的 Sm、Nd 含量分别为 $5.05 \times 10^{-6} \sim 5.51 \times 10^{-6}$ 和 $23 \times 10^{-6} \sim 25 \times 10^{-6}$, $^{147}\text{Sm}/^{144}\text{Nd}$ 值变化范围较大 (0.1305 ~ 0.1333), $^{143}\text{Nd}/^{144}\text{Nd}$ 值范围比较集中, 为 0.512319 ~ 0.512384, $\epsilon_{\text{Nd}}(0)$ 值范围为 -6.2 ~ -5.0, $\epsilon_{\text{Nd}}(t)$ 值范围较

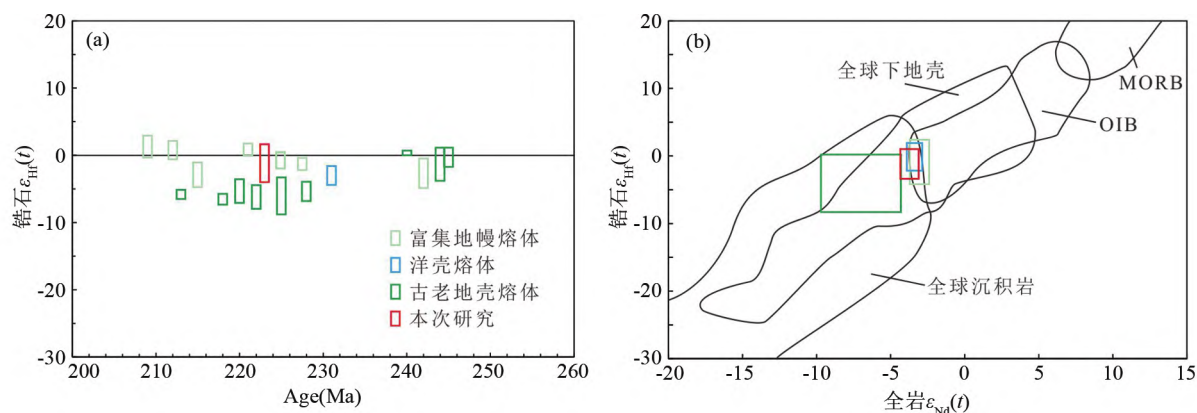


图9 夏日哈木安山岩 Nd-Hf 同位素图解

(a) 锆石年龄- $\epsilon_{\text{Hf}}(t)$ 图, 东昆仑地区富集地幔、洋壳和古老地壳熔体推测数据: Ding *et al.* (2014), Li *et al.* (2015), Yin *et al.* (2017), Zhao *et al.* (2020), Fan *et al.* (2022), Chen *et al.* (2022), Zhu *et al.* (2022); (b) 全岩 $\epsilon_{\text{Nd}}(t)$ -锆石 $\epsilon_{\text{Hf}}(t)$ 图, OIB 和 MORB: Kempton and McGill (2002); 全球沉积岩: Vervoort and Blichert-Toft (1999); 全球下地壳: Dobosi *et al.* (2003)

Fig. 9 Nd-Hf isotopic diagrams for the Xiarihamu andesites

(a) zircon ages vs. $\epsilon_{\text{Hf}}(t)$ plot, fields of supposed enriched mantle, oceanic crust and ancient continental crust data of East Kunlun from Ding *et al.* (2014), Li *et al.* (2015), Yin *et al.* (2017), Zhao *et al.* (2020), Fan *et al.* (2022), Chen *et al.* (2022), Zhu *et al.* (2022); (b) whole rock $\epsilon_{\text{Nd}}(t)$ vs. zircon $\epsilon_{\text{Hf}}(t)$ plot, fields of OIB and MORB from Kempton and McGill (2002); Global sediments from Vervoort and Blichert-Toft (1999); Lower crust from Dobosi *et al.* (2003)

小, 为 $-4.3 \sim -3.1$ 。

4.3 锆石 U-Pb 年代学

采用 LA-ICP-MS 法对安山岩 (K17-5-8.9) 中选取的 16 颗锆石进行了测试 (表 3)。根据锆石阴极发光图像 (图 6), 锆石颗粒自形程度较高, 颗粒大小约 $100 \sim 150 \mu\text{m}$, 多呈短柱状, 围绕核部具有明显的震荡环带, 个别发育扇形分带或平行晶体长轴的生长条带; 少部分锆石边部具有溶蚀结构, 反映后期可能受到改造作用。锆石 Th/U 比值为 $0.58 \sim 1.31$ 之间, 平均 0.81, 具岩浆锆石特征。锆石年龄集中分布在谐和线上 (图 7), $^{206}\text{Pb}/^{238}\text{U}$ 加权平均年龄 $223 \pm 2 \text{ Ma}$ (MSWD = 1.5, $n = 16$) 代表了锆石的结晶年龄, 指示安山岩的形成时代为晚三叠世。

4.4 锆石 Lu-Hf 同位素特征

在安山岩中选定的锆石颗粒上进行 Lu-Hf 同位素测试 (表 4)。锆石 $^{176}\text{Yb}/^{177}\text{Hf}$ 值范围从 0.0257 到 0.0685, $^{176}\text{Lu}/^{177}\text{Hf}$ 值范围从 0.0006 到 0.0014。所有测点都具有小于 0.002 的 $^{176}\text{Lu}/^{177}\text{Hf}$ 值, 表明放射性成因 ^{176}Hf 很少。 $^{176}\text{Hf}/^{177}\text{Hf}$ 范围在 0.282540 到 0.282668 之间, $\epsilon_{\text{Hf}}(t)$ 值范围从 -3.4 到 $+1.1$, 加权平均数为 -1.3 。锆石模式年龄 t_{DM} 在 $831 \sim 1003 \text{ Ma}$ 之间, t_{DM2} 在 $1591 \sim 1994 \text{ Ma}$ 之间。

5 讨论

5.1 岩石成因

本次研究的样品中除两个样品外其余样品烧失量 (LOI)

皆小于 2.1% (表 1), 表明岩石蚀变较弱。从岩石化学特征看, 本次研究的火山岩岩石类型以安山岩为主, 未见玄武岩出露, 岩石地球化学成分均一。考虑到岩石锆石中未发现继承锆石, 且具有相对稳定的全岩 $\epsilon_{\text{Nd}}(t)$ 值和锆石 $\epsilon_{\text{Hf}}(t)$ 值, 稀土、微量元素图解中各样品分配形式一致 (图 5), 表明岩石保留了其原始地球化学性质, 且 Nb、Th、Yb、Nd、Sm、La、Ba 等元素在化学风化作用中较稳定, 可用于讨论安山岩的成因。

安山岩 La/Yb-La 图解 (图 8a) 表明岩石主要为部分熔融形成, 分离结晶作用并不明显。安山岩具有较低的 SiO_2 ($59.45\% \sim 63.10\%$)、较高的 Al_2O_3 ($16.19\% \sim 17.66\%$) 含量特征, 结合较低的 Co ($6.72 \times 10^{-6} \sim 16.5 \times 10^{-6}$)、Ni ($4.58 \times 10^{-6} \sim 10.4 \times 10^{-6}$) 含量特征表明岩石可能起源于地壳熔体。研究表明, 由加厚下地壳拆沉部分熔融产生的岩浆岩具有较高 (>45) 的 $\text{Mg}^\#$ (Rapp and Watson, 1995), 本次研究的样品 $\text{Mg}^\#$ 值介于 $42 \sim 51$ (其中 1 个样品为 28, 可能由蚀变引起), 13 个样品中有 9 个样品 $\text{Mg}^\#$ 值为 $45 \sim 51$, 较高的 $\text{Mg}^\#$ 值以及 La/Yb 和 La 相关关系表明其主要为地壳物质的部分熔融, 受到地幔熔体的影响 (图 8b), 西部样品 (图 3d) 中的斜长石具有环带结构也可能与岩浆混合作用有关。样品的 Th 含量 ($11.0 \times 10^{-6} \sim 15.1 \times 10^{-6}$, 平均 12.4×10^{-6}) 和 U 含量 ($2.51 \times 10^{-6} \sim 3.98 \times 10^{-6}$, 平均 3.04×10^{-6}) 高于平均上地壳值 (10.5×10^{-6} 和 2.7×10^{-6} , Rudnick and Gao, 2003), 暗示其不可能为地幔源区部分熔融导致。样品的 La/Nb 值为 $3.02 \sim 3.51$, 高于平均地壳 (2.2) 和地幔 (1) 组成, Nb/Ta 值为 $11.85 \sim 17.46$, 高于平均地壳 (11) 但略小于地幔 (17.7) 值, 同样表明岩石主要为壳源成因, 但受到壳幔混合

作用的影响。样品具有相对较低的 K_2O/Na_2O 、 $Al_2O_3/(MgO + TiO_2 + Fe_2O_3^T)$, 暗示岩石可能源于下地壳的脱水熔融(图 8c)。Rapp *et al.* (1991) 实验岩石学研究表明, 玄武岩在 0.8 GPa 下发生部分熔融时残留相为角闪石 + 斜长石 ± 斜方辉石(无石榴石), 在 1.6 GPa 时, 残留相为石榴石 + 单斜辉石 ± 角闪石 ± 斜长石 ± 斜方辉石。岩石具有较低压力熔融的主量元素特征(图 8c), 同时具有负 Eu 异常特征, 结合岩石具有较低的 Sr/Y、La/Y 值(图 8d), 源区可能残留斜长石, 推测岩石源于较低压力条件下的部分熔融。

样品属于高钾钙碱性系列, 具有富集 LILEs 和亏损 Nb、Ta 的岛弧相关岩浆特征, 可能反映了陆壳与地幔的混合过程。锆石 $\varepsilon_{Hf}(t)$ 值大于 0 表明岩浆源区为亏损地幔或新生地壳, 反之则以古老地壳或富集地幔源区为主体(Vervoort and Patchett, 1996), 安山岩锆石 $\varepsilon_{Hf}(t)$ 值较低(-3.4 ~ +1.1), 主体小于 0, 表明其主要来源于古老地壳或富集地幔源区。由于样品锆石普遍具有高于古老地壳熔体锆石 Hf 同位素的特征, 并且与富集地幔源区的熔体相近(图 9a), 结合岩石地壳化学特征, 表明其可能来源于古老下地壳部分熔融, 并受富集地幔物质的影响(图 8c)。同时, 全岩 Sm-Nd 同位素模式年龄(1418 ~ 1538 Ma), 锆石 Lu-Hf 同位素一阶段模式年龄(t_{DM}) 在 831 ~ 1003 Ma 之间, 二阶段模式年龄(t_{DM2}) 在 1591 ~ 1994 Ma 之间, 远大于其形成年龄, 这与鄂拉山组流纹岩具有相同的特征(Fan *et al.*, 2022; 封铨等, 2022), 与东昆仑古老基底金水口群的形成年代一致(He *et al.*, 2016)。结合主量元素特征(图 8c), 安山岩可能来源于中-新元古代金水口群中的斜长角闪岩。夏日哈木安山岩 $\varepsilon_{Nd}(t)$ 值(-4.3 ~ -3.1) 位于古老地壳熔体与富集地幔熔体之间(图 9b), 与东昆仑地区亮源 I 型花岗岩(-10.5 ~ -3.1, Shao *et al.*, 2015; Xiong *et al.*, 2016) 相近, 略低于来源于富集地幔熔融的中基性岩(-3.9 ~ +0.1, Hu *et al.*, 2016; Liu *et al.*, 2017; Yin *et al.*, 2017; Zhao *et al.*, 2020), 远低于亏损地幔来源的 N-MORB(+12.2 ~ +12.8, 郭安林等, 2007)。同时, 岩石落入全球下地壳熔体范围(图 9b), 进一步暗示岩石可能起源于下地壳熔体。

5.2 构造意义

本次研究的安山岩样品形成年龄为 223 Ma, 属于晚三叠世(图 7), 与祁漫塔格野牛沟、土房子地区基性岩墙和火山岩时代相近, 后者年龄集中在 220 ~ 218 Ma(Hu *et al.*, 2016)。前人将东昆仑地区广泛分布的一套晚三叠世中-酸性钙碱性系列火山岩组合归属为鄂拉山组(八宝山组; 姜春发等, 1992)。相比东昆仑西段而言, 东昆仑东段鄂拉山组火山岩的研究程度较高, 都兰-香日德地区鄂拉山组火山岩形成时代集中在 235 ~ 210 Ma, 如波洛斯英安岩(235 ~ 233 Ma; 封铨等, 2022)、鄂拉山流纹岩(228 Ma; Hu *et al.*, 2016)、哈日扎英安岩(228 ~ 225 Ma; Fan *et al.*, 2022)、那更英安岩(222 ~ 218 Ma; Fan *et al.*, 2022) 和流纹岩(228 Ma、

213 Ma; Chen *et al.*, 2022)、香日德流纹岩(214 ~ 212 Ma; 丁烁等, 2011) 等。晚三叠世东昆仑不同地区可能存在板片脱离、岩石圈地幔拆沉、软流圈地幔上涌等不同作用(Fan *et al.*, 2022), 表明不同地区的鄂拉山组火山岩可能为不同构造-岩浆事件的产物。

区域地质研究表明, 布青山蛇绿混杂岩所代表的阿尼玛卿-布青山洋是东昆仑造山带晚古生代时期的主洋盆, 控制着古特提斯洋演化历史和区域构造格架(姜春发等, 1992)。尽管古特提斯洋打开的精确时限尚不清楚, 但近年来有学者认为古特提斯洋在早泥盆世已经开始打开, 早石炭世已成为成熟大洋(Dong *et al.*, 2020; 李文渊等, 2022a, b), 也有学者认为原-古特提斯洋构造体制转换发生在晚泥盆世(杨金中等, 1999; Wang *et al.*, 2022b), 不过普遍认为古特提斯洋在晚石炭世处于扩张状态(Yang *et al.*, 2009; 刘战庆等, 2011; Wang *et al.*, 2022a), 随后大洋持续俯冲消减直至闭合。对古特提斯洋闭合的精确时限也存在很多不同观点, Yang *et al.* (2009) 认为早-中二叠世大洋已经关闭; 吴福元等(2020) 认为大洋在早三叠世(250 Ma) 关闭; 熊富浩等(2011) 认为早三叠世(251 Ma) 东昆仑仍处于俯冲环境, 具有典型的沟弧盆体系, 其中, 中灶火和巴隆地区发育早三叠世弧后伸展背景下的镁铁质岩墙群(259 ~ 249 Ma)(罗文行等, 2013; 菅坤坤等, 2015; Xiong *et al.*, 2019); Zhao *et al.* (2020) 认为中三叠世(240 Ma) 东昆仑由俯冲转为碰撞环境; 此外, 也有学者认为古特提斯洋的俯冲过程至少持续到中三叠世晚期(Zhang *et al.*, 2012; Xiong *et al.*, 2014, 2016; Li *et al.*, 2018b)。中-晚三叠世(~240 Ma) 古特提斯洋进入碰撞造山时期, 形成一系列同碰撞侵入岩和火山岩(Huang *et al.*, 2014; Shao *et al.*, 2017)。

晚三叠世侵入岩(230 ~ 223 Ma) 和成矿峰期表明该时期可能为古特提斯洋构造由挤压向伸展转换的关键时期(Xiong *et al.*, 2014; Ren *et al.*, 2016; 陈国超等, 2020; Zhao *et al.*, 2020), 这也得到很多学者的支持。Hu *et al.* (2016) 认为晚三叠世基性岩墙群(228 ~ 218 Ma) 是碰撞后伸展背景的产物; Dong *et al.* (2018) 认为 220 ~ 200 Ma 一系列蛇绿混杂岩相关的侵入岩、火山岩和沉积序列代表碰撞后伸展环境的产物; Zhang *et al.* (2017) 认为东昆仑东段赛什堂地区富 Cu 斑岩(218 ~ 224 Ma) 来源于碰撞后伸展背景下的岩浆活动。与俯冲和同碰撞背景不同, 这一时期主要形成具有后碰撞特征的 I 型花岗岩、埃达克质和少量 A2 型花岗岩(Hu *et al.*, 2016; Kamaunji *et al.*, 2021; Fan *et al.*, 2022), 岩浆向高钾钙碱性系列演化, 部分岩浆岩具 A 型花岗岩和埃达克质岩岩浆特征也反映了东昆仑地区在 220 ~ 210 Ma 已处于加厚下地壳拆沉导致的伸展构造背景中(Shao *et al.*, 2021; Zhu *et al.*, 2022)。虽然从碰撞挤压到碰撞后延伸的确切时间仍存在争议, 但区域地质研究表明在 ~230 Ma, 东昆仑构造体制已由碰撞挤压向伸展环境转变。另外, 沉积岩方面的证据表明, 下三叠统洪水川组、中三叠统闹仓沟组发

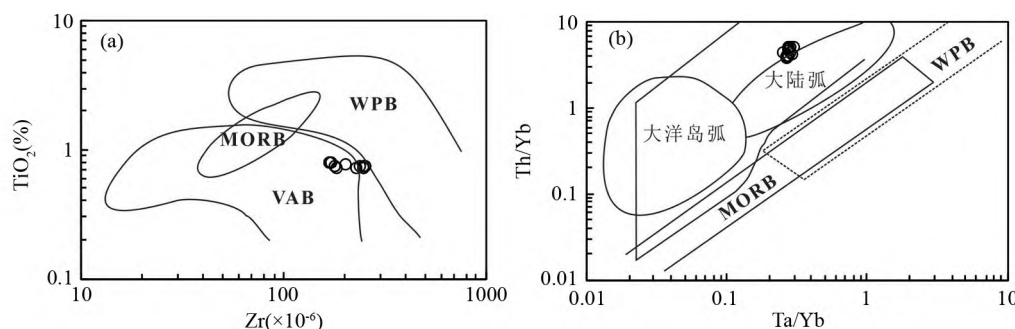


图10 夏日哈木安山岩构造环境判别图(据Pearce, 1982)

(a) TiO_2 -Zr 图解; (b) Th/Yb-Ta/Yb 图解

Fig. 10 Tectonic discrimination diagrams of Xiarihamu andesites (after Pearce, 1982)

育弧前盆地沉积体系,其中的英安质火山岩具有活动大陆边缘岛弧性质(Li *et al.*, 2015),中三叠统希里克特组具有海陆交互沉积特征,上三叠统鄂拉山组(八宝山组)为陆相碎屑岩沉积组合(姜高磊等, 2014),反映了古特提斯洋晚三叠世早期大陆碰撞基本完成(李瑞保等, 2012)。

本次安山岩样品显示出贫铁、镁而富碱的特征,稀土配分型式呈向右缓倾的轻稀土富集型,但重稀土分馏不明显,具相对平坦的型式,亏损亏损;各稀土曲线近于平行,其分布型式与现代岛弧火山岩稀土元素配分型式相似,反映岩浆来源于轻稀土未亏损的或轻微富集的源区,与东昆仑造山带三叠纪(250~235Ma)同碰撞花岗岩明显的负Eu异常(图5)有所区别(Xia *et al.*, 2014; Li *et al.*, 2015; Shao *et al.*, 2017),暗示安山岩与三叠纪同碰撞花岗岩俯冲洋壳部分熔融成因有所不同(Niu *et al.*, 2013)。微量元素特征表现为强不相容元素Rb、Ba、U、Th强烈富集,Ta、Nb、Ti亏损,样品的Zr/Y值为6.33~8.65,介于大陆弧范围内(4~12; Condie, 1989),La/Nb比值(3.02~3.36)也与活动大陆边缘弧岩浆特征一致(大于2)。全球新生代安山岩地球化学成分研究显示,常用的玄武岩构造环境判别图解可用于安山岩环境判别中(刘欣雨等, 2017), TiO_2 -Zr图解(图10a)中岩石落入火山弧玄武岩区,在Th/Yb-Ta/Yb图解(图10b)中落入大陆弧区,同时安山岩 $\varepsilon_{\text{Nd}}(t)$ 值和锆石 $\varepsilon_{\text{Hf}}(t)$ 值也与东昆仑地区中三叠世岛弧花岗岩相似(Li *et al.*, 2018a),说明形成安山岩的源区与岛弧岩浆有关。通常认为,大洋俯冲消减作用会形成大量具有陆缘弧性质的中酸性侵入岩、火山岩以及弧后盆地岩浆岩组合(莫宣学等, 2007),如巴隆地区和哈拉尔吐地区的岛弧花岗岩(263~244Ma; Zhang *et al.*, 2012; Li *et al.*, 2018a);沟里地区和大灶火沟的岛弧火山岩(255~244Ma; 史连昌等, 2016; 张新远等, 2020)。但近年来有学者在东昆仑西段识别出晚三叠世岛弧岩浆岩(邓红宾等, 2018; 徐博等, 2020),据此提出古特提斯洋在东昆仑西段关闭时间较晚,可推迟到晚三叠世之后(邓红宾等, 2018),认为晚三叠世晚期仍存在洋壳俯冲消减作用(刘红涛, 2001)。实际上,具备岛弧特征的岩浆岩并不一定代表其形成时存在

俯冲作用,其源区的“富集”特征可能是由早期的俯冲作用造成的。如在东昆仑东段晚三叠世晚期(216Ma)同样可形成岛弧特征的岩浆岩(张明东等, 2018)。

本次研究的安山岩属于高钾钙碱性系列岩石,一般认为,该系列岩石形成于造山后伸展背景环境(邓晋福等, 2004)。碰撞伸展环境的岩浆岩源区会受到早期洋/陆俯冲作用的影响,导致其表现出类似岛弧火山岩特征,如富集大离子亲石元素和轻稀土元素,亏损高场强元素和Nb、Ta,造山带中板块俯冲的岩浆作用可能存在延迟效应(张明东等, 2018)。因此,碰撞后伸展环境下也可以产生具有岛弧特征的岩石(Aldanmaz *et al.*, 2000)。东昆仑地区发育的岛弧性质中酸性火山岩具有低的 $\varepsilon_{\text{Nd}}(t)$ 、较高的Nb/Ta值和Zr/Hf值,常被认为与壳幔混合作用有关(Xiong *et al.*, 2012),罗照华等(2002)认为幔源玄武质岩浆底侵作用可能从早三叠世(~250Ma)一直持续到晚三叠世(~220Ma),导致碰撞后伸展环境产生“俯冲”印记的岩浆岩,在时间上产生了滞后效应。夏日哈木地区的晚三叠世安山岩就是这种滞后效应在东昆仑西段的体现。

综上所述,古特提斯洋在晚三叠世已进入碰撞后伸展环境,夏日哈木地区晚三叠世(223Ma)安山岩虽然具有“岛弧”特征,但形成于古特提斯洋闭合之后。碰撞后的伸展作用引发早期俯冲作用有关的地幔源区发生部分熔融,形成的玄武质岩浆底侵,造成古老下地壳熔融,最终形成夏日哈木安山岩。

6 结论

(1) 夏日哈木地区鄂拉山组安山岩 LA-ICP-MS 锆石 U-Pb 年龄为 223 ± 2 Ma,表明其形成于晚三叠世。

(2) 安山岩属于高钾钙碱性岩石系列,具有富集轻稀土、大离子亲石元素,亏损高场强元素的岛弧火山岩特征,全岩 $\varepsilon_{\text{Nd}}(t)$ 值(-4.3~-3.1)和锆石 $\varepsilon_{\text{Hf}}(t)$ 值(-3.4~+1.1)表明岩石来源于受富集地幔熔体底侵影响的古老下地壳部分熔融体。

(3) 与古特提斯洋演化有关的晚三叠世安山岩形成于碰撞后伸展环境。

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